Diagrams of Thermodynamic State with MatLab

Fabian Wermelinger *
Lucerne University
Mechanical Engineering

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Abstract

The following text is a manual to the function `thermo_diagram_plot.m` for use in a MatLab environment. The function can be used to plot diagrams of thermodynamic state of water. You can choose between three types of diagrams, which are a T, s, h, s and $\log p, h$–Diagram.

Inhaltsverzeichnis

* fabian.wermelinger@stud.hslu.ch

1 Dependencies

All calculations are made with help of `XSteam.m`. The file is freely available at `http://www.mathworks.com/matlabcentral/fileexchange/9817` and was developed by Magnus Holmgren (`http://www.x-eng.com`). The calculations follow the IAPWS IF-97 standard. `XSteam.m` is already integrated into `thermo_diagram_plot.m`. Therefore, you must not take care of that file at MatlabCentral.

For the sake of operational speed, `thermo_diagram_plot.m` makes use of a file called

```
wet_steam_area_*.mat
```

in the working directory[1]. If there is no `.mat`-file present, switch the `'WetSteamArea'` option to `'On'` and the file will be created. More on this option, see Section 3.4 on page 5.

2 Syntax

The function uses the following basic syntax

```
[out1,out2] = thermo_diagram_plot('type',in1,[in2],['option'],['value']);
```

Variables in brackets are optional. Two output arguments `out1` and `out2` can be created, while `out1` is created anyway.

2.1 Input Arguments

`thermo_diagram_plot.m` needs at least two input arguments. The first (`'type'`) determines the diagram to be plot and must be passed as a *string*[2]. The following table shows the relation between the available diagrams and their strings:

Diagram type	String (`'type'`)
T, s–Diagram	`'Ts'`
h, s–Diagram	`'hs'`
$\log p, h$–Diagram	`'logph'`

`thermo_diagram_plot.m` is not *Case-Sensitive*, that is, lower case and upper case letters do not matter.

The second mandatory input argument `in1` is dependent on the desired diagram. For a T, s and h, s–Diagram's, `in1` must be a pressure p. For a $\log p, h$–Diagram, `in1` must be a temperature T. The variable `in1` can be either a scalar or vector. If it is a scalar k, then the iso-curve $f_k = f(k)$ with the iso-value k will be plotted. If `in1` is a $[1 \times n]$ or $[n \times 1]$ vector with elements a_i for $i = 1 \ldots n$, then n iso-curves $f_i = f(a_i)$ with iso-values a_i will be plotted[3].

[1] * is called a "Wildcard" and is used as a place holder for any additional characters.
[2] Strings must be passed as characters between inverted comma.
[3] Either row or column vectors can be passed to the function.

Input argument `in2` is optional. The argument defines the range over which the diagram will be plot. If it is omitted, a default range will be assigned. For the T, s and h, s–Diagram's, `in2` defines the *entropy range* and for the $\log p, h$–Diagram the *pressure range*. The default for the entropy range is

$$0\,\mathrm{kJ/(kgK)} \leq s \leq 10\,\mathrm{kJ/(kgK)}$$

and the pressure range

$$0.00612\,\mathrm{bar} \leq p \leq 1000\,\mathrm{bar}$$

For a user defined range, `in2` must be a vector. Take care for a fine discretization if you want accurate results. Especially the $\log p, h$–Diagram is very sensitive in that case.

If `in1` and `in2` are both passed as *scalars*, the single point $R = R(\mathtt{in1}, \mathtt{in2})$ will be plotted only. To plot multiple points, see Section 3.3 on page 5.

The following units must be hold for input arguments

$$[T] = {}^\circ\mathrm{C}$$
$$[s] = \mathrm{kJ/(kgK)}$$
$$[p] = \mathrm{bar}$$

Additional input arguments can be passed as options. All options will be further discussed in Section 3.

2.2 Output Arguments

If you need the numerical values of the plotted data, you can do so by defining output arguments. Refer to the basic syntax to see how or check out example 4.1 on page 6).

The output `out1` is dependent on the dimension of `in1`. This output is a column vector if `in1` is a scalar or a $[m \times n]$ matrix if `in1` is a vector with n elements. Any column of `out1` correspond to the values of an iso-line, for which `in1` defines the iso-value. The output `out1` are the y-values of the plots.

The second output `out2` are the x-values of the plots. If the input argument `in2` was defined, then the output `out2` is identical to that argument, otherwise the variable contains the default values.

The $\log p, h$–Diagram has a special bevahior considering the output arguments. For this diagram the dimensions of the output variables `out1` and `out2` are switched. The definition of the value assignment will be the same thought.

Output arguments have following units:

$$[T] = {}^\circ\mathrm{C}$$
$$[s] = \mathrm{kJ/(kgK)}$$
$$[p] = \mathrm{bar}$$
$$[h] = \mathrm{kJ/kg}$$

3 Options

You can pass additional options to the functions, which are in given in the common MatLab style[4]. The option pairs (identifier and value) are separated by a comma. Where the identifier is always a string, the value can be both a string or a numerical value. The order of the option pairs (`'identifier', 'value'`) nor case sensitivity does not matter.

The following options are valid for *all* diagram types of `thermo_diagram_plot.m`.

3.1 Constant Vapor Fraction Lines

The boiling and dew curves are always displayed by default and can not be altered. However, vapor fraction lines can be displayed additionally by passing this option.

```
'VaporFractions'
```

with the possible values

```
'On'
'Off'
```

The default is `'Off'`. Refer to example 4.1 on page 6.

3.2 Axes Limits

For a better close up, user defined axes limits can be passed to the function. This option is identical to the `XLim` and `YLim` properties of an axes object in MatLab.

```
'XLim'
'YLim'
```

The possible value is a two element vector. The first element defines the minimum value, where the second element defines the maximum value of the according axes. The minimum value must be *less* than the maximum.

```
[minimum maximum]
```

The YLim option for a $\log p, h$–Diagram is special in this case again. Due to the layout of the plot, this logarithmically spaced axis can only take values which are powers of ten! The following list shows all possible values for the `minimum` and `maximum` variables in this special case:

$$10^{-2} \quad 10^{-1} \quad 10^{0} \quad 10^{1} \quad 10^{2} \quad 10^{3}$$

[4]Options are passed in MatLab by a leading identifier and followed by the value the option has to be assigned with. The same convention is used here.

3.3 Multiple Points

Without this option the function plots a single point if `in1` and `in2` are scalars. If you want to plot multiple single points at one time you need to declare this option.

```
'MultiplePoints'
```

with the possible values

```
'On'
'Off'
```

The default is `'Off'`.

If you choose this option, the input arguments `in1` and `in2` must be vectors with n elements a_i and b_i. By doing so you get n points $R_i = R(a_i, b_i)$ for $i = 1 \ldots n$. Check out example 4.2 on page 6 for further insight.

3.4 Create an Outsourcing File

As mentioned in Section 1 the function uses an outsource for faster calculations. The function assumes that the file is present in the working directory. If not, you can create one by passing this option to the function.

```
'WetSteamArea'
```

with the possible values

```
'On'
'Off'
```

The default is `'Off'`.

3.5 Print Hard-Copies

The generated plot by calling the function is automatically saved in a `.pdf` file to the working directory. By default this is a landscape A4 paper size. Passing this option means to alter paper size to most of MatLab's known formats.

```
'Print'
```

with the possible values

```
'A0', 'A1', 'A2', 'A3', 'A4', 'A5',
'B0', 'B1', 'B2', 'B3', 'B4', 'B5',
'usletter', 'uslegal', 'tabloid'
'arch-A', 'arch-B', 'arch-C', 'arch-D', 'arch-E',
'A', 'B', 'C', 'D', 'E'
```

3.6 Switch the Legend On and Off

By default a legend is created with any plot. If you feel it is disturbing to show a legend, simply switch it off with this option.

```
'Legend'
```

with the possible values

```
'On'
'Off'
```

The default is `'On'`.

4 Examples

4.1 Constant Vapor Fraction Lines

The following example creates a T, s–Diagram with three isobaric lines for $p_1 = 1\,\text{bar}$, $p_2 = 200\,\text{bar}$ and $p_3 = 500\,\text{bar}$. Additionally, vapor fraction lines of constant value shall be displayed.

MatLab prompt:

```
>> p = [1 200 500];
>> [a,b] = thermo_diagram_plot('Ts',p,'VaporFractions','On');
```

In this example, the numerical output is wished to be saved in the variables a and b. If we take a look at the workspace we can examine the dimensions of the single matrices and vectors.

```
>> whos
Name          Size                 Bytes  Class     Attributes

a          1001x3                  24024  double
b             1x1001                8008  double
p             1x3                     24  double
```

There are three iso-values saved in the variable p. Therefore, the matrix a containing the temperature values of the iso-lines has three columns at $m = 1001$ rows. The rows determine the number of values in x direction, which is entropy in this case saved in variable b.

You can find the plot on page 8.

4.2 Multiple Points

The following example creates a h, s–Diagram with three single points at $p_1 = 1\,\text{bar}$, $p_2 = 200\,\text{bar}$ and $p_3 = 500\,\text{bar}$ as well as $s_1 = 2\,\text{kJ/(kgK)}$, $s_2 = 7\,\text{kJ/(kgK)}$ and $s_3 = 5\,\text{kJ/(kgK)}$. There is a user defined range given as well.

MatLab prompt:

```
>> p = [1 200 500];
>> s = [2 7 5];
>> thermo_diagram_plot('hs',p,s,'MultiplePoints','On', ...
                         'XLim',[1 8],'YLim',[500 4100]);
```

No numerical values are written to the workspace in this case.

You can find the plot on page 9.

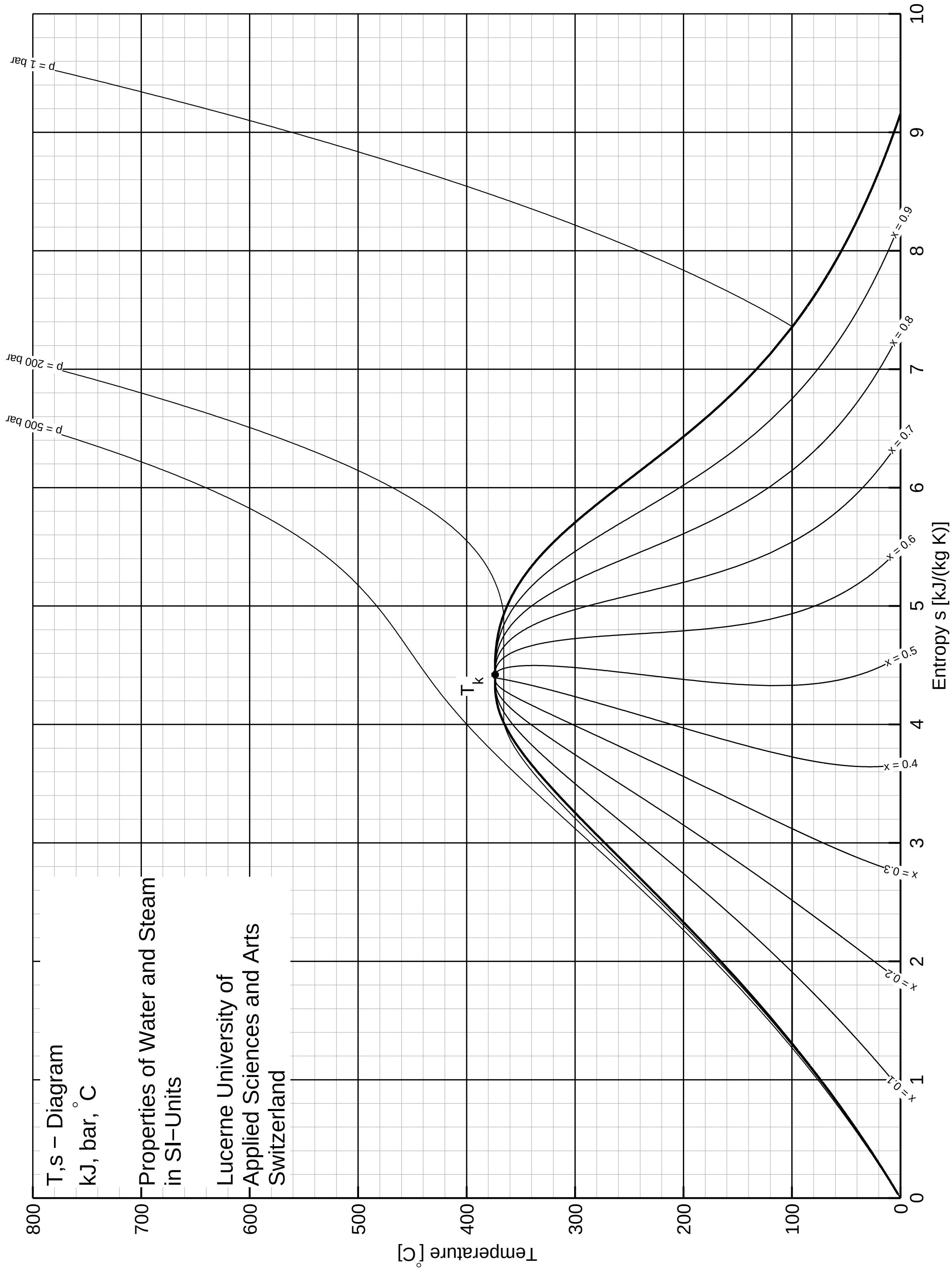
4.3 $\log p, h$–Diagram

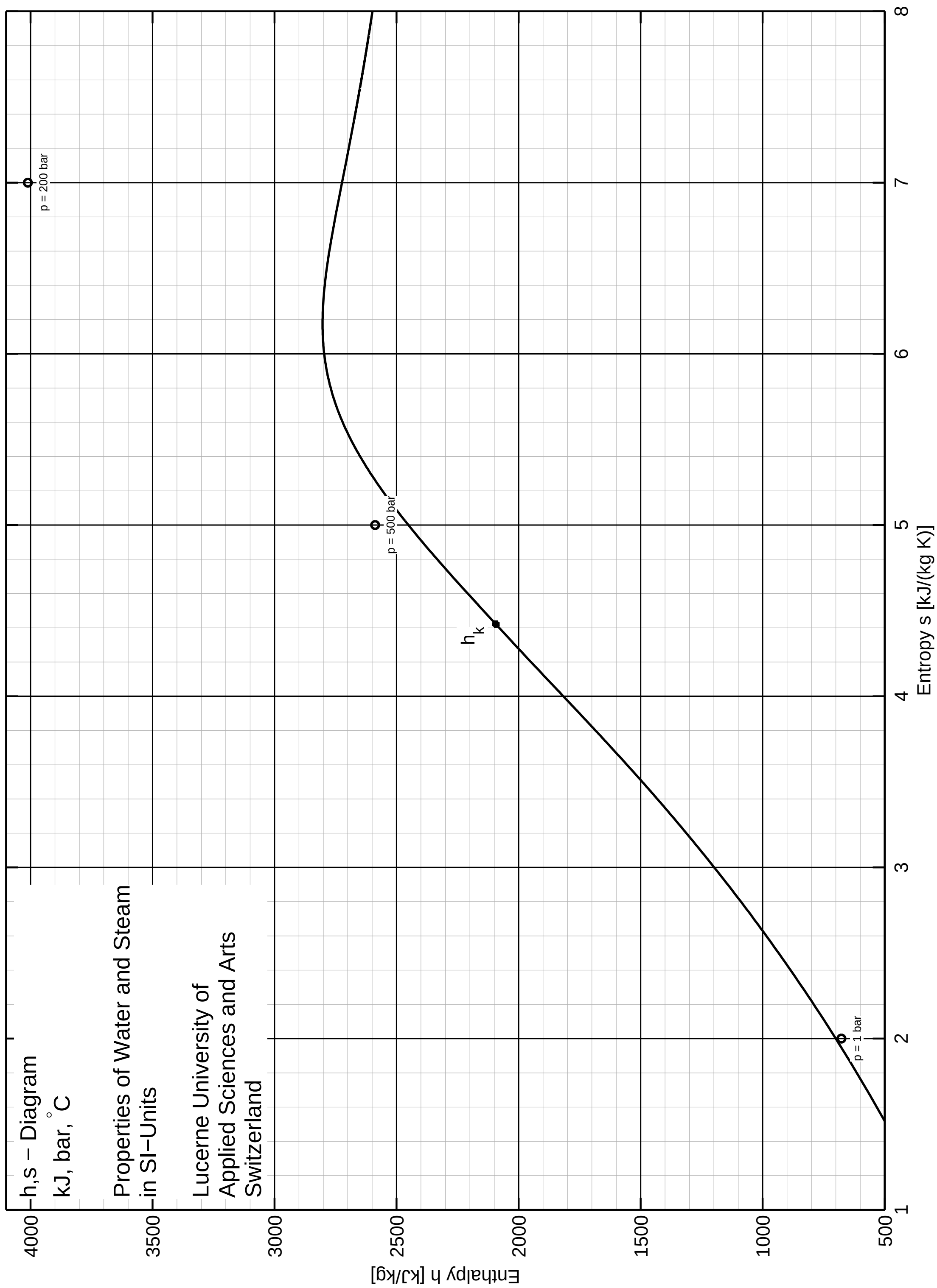
The following example creates a $\log p, h$–Diagram with three isotherm $T_1 = 20\,^{circ}$C, $T_2 = 200\,^{circ}$C and $T_3 = 800\,^{circ}$C. Some options are passed as well.

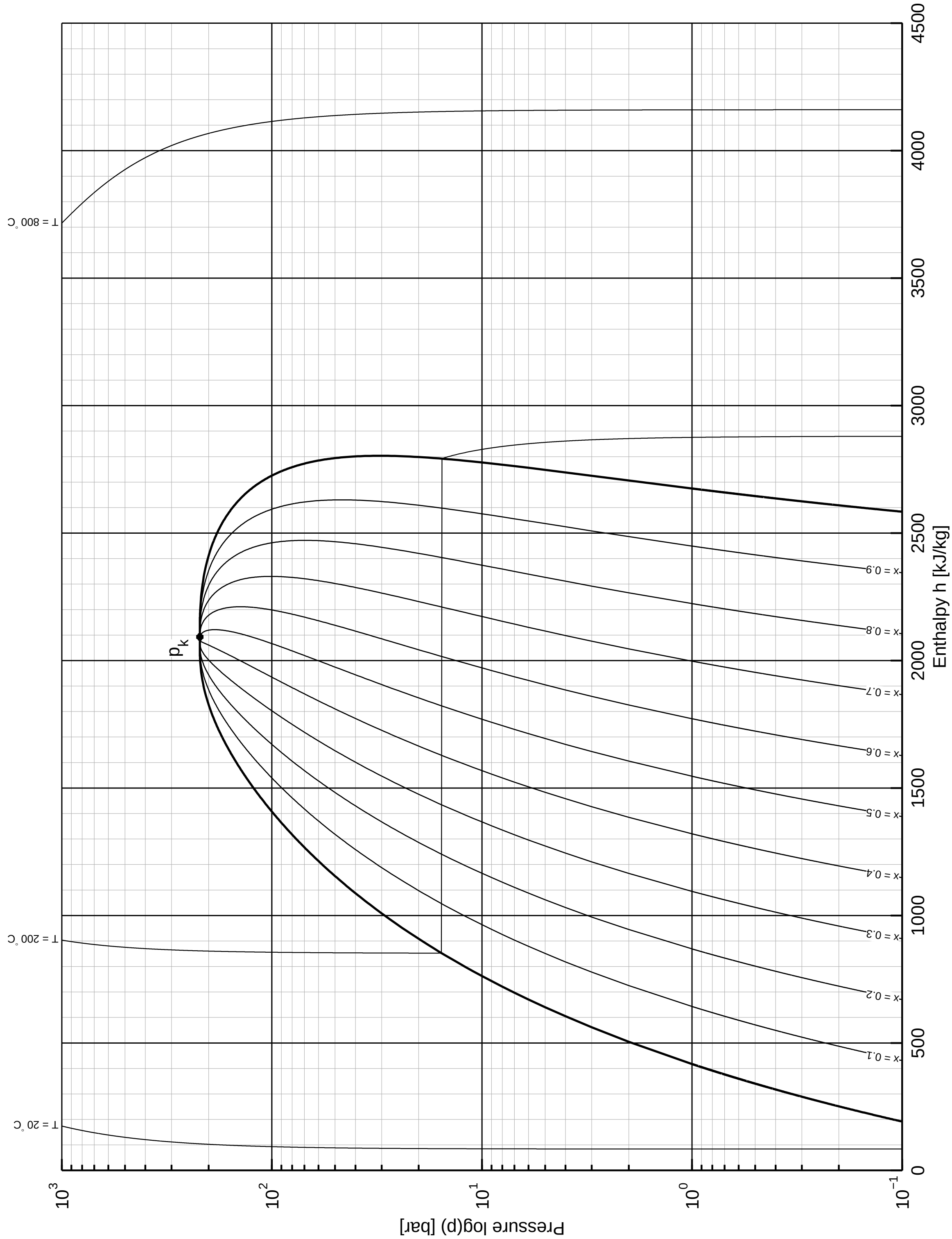
MatLab prompt:

```
>> T = [20 200 800];
>> thermo_diagram_plot('logph',T,'YLim',[10^-1 10^3], ...
                         'VaporFractions','on','Legend','off');
```

You can find the plot on page 10.


T,s – Diagram
kJ, bar, °C
Properties of Water and Steam
in SI–Units
Lucerne University of
Applied Sciences and Arts
Switzerland
p = 1 bar
p = 200 bar
p = 500 bar
x = 0.1
x = 0.2
x = 0.3
x = 0.4
x = 0.5
x = 0.6
x = 0.7
x = 0.8
x = 0.9
T_k
Temperature [°C]
Entropy s [kJ/(kg K)]
0
100
200
300
400
500
600
700
800
0
1
2
3
4
5
6
7
8
9
10


h,s – Diagram
kJ, bar, °C
Properties of Water and Steam
in SI–Units
Lucerne University of
Applied Sciences and Arts
Switzerland
p = 200 bar
p = 500 bar
p = 1 bar
h_k
Entropy s [kJ/(kg K)]
Enthalpy h [kJ/kg]
1
2
3
4
5
6
7
8
500
1000
1500
2000
2500
3000
3500
4000


Enthalpy h [kJ/kg]
Pressure log(p) [bar]
0
500
1000
1500
2000
2500
3000
3500
4000
4500
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
T = 20 °C
T = 200 °C
T = 800 °C
p_k
x = 0.1
x = 0.2
x = 0.3
x = 0.4
x = 0.5
x = 0.6
x = 0.7
x = 0.8
x = 0.9